

HW Class 6 (R functions)

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```
library(bio3d)
```

```
# reads PDB file, extracts CA atoms from selected chain, gets B-factors, and plots them
analyze_protein <- function(pdb_id, chain_id = "A") {

  # read PDB structure from database
  pdb <- read.pdb(pdb_id)

  # keep only CA atoms from specified chain
  pdb_chain <- trim.pdb(pdb, chain = chain_id, eley = "CA")

  # extract B-factor values
  b_factors <- pdb_chain$atom$b

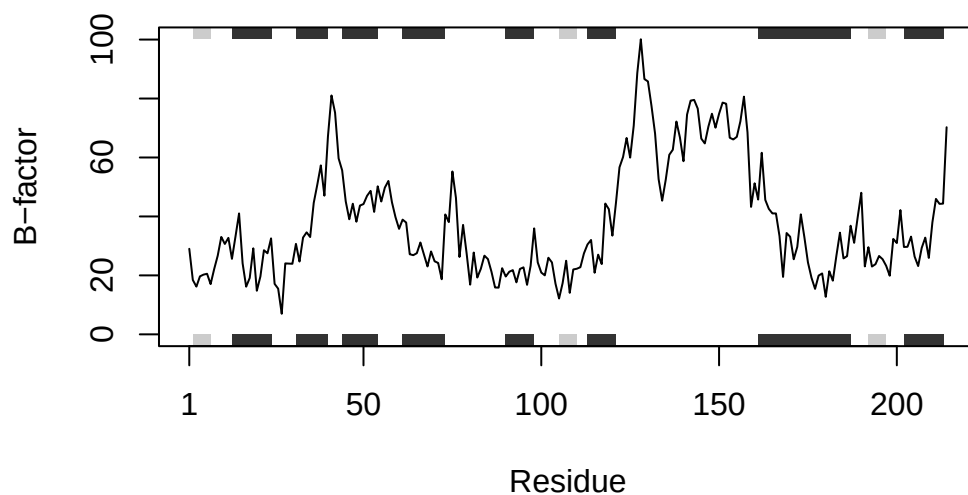
  # plot B-factors
  plotb3(b_factors, sse = pdb_chain, typ = "l",
         ylab = "B-factor",
         main = paste("B-factor Plot for", pdb_id))

  # return B-factor vector
  return(b_factors)
}
```

```
analyze_protein("4AKE")
```

Note: Accessing on-line PDB file

B-factor Plot for 4AKE



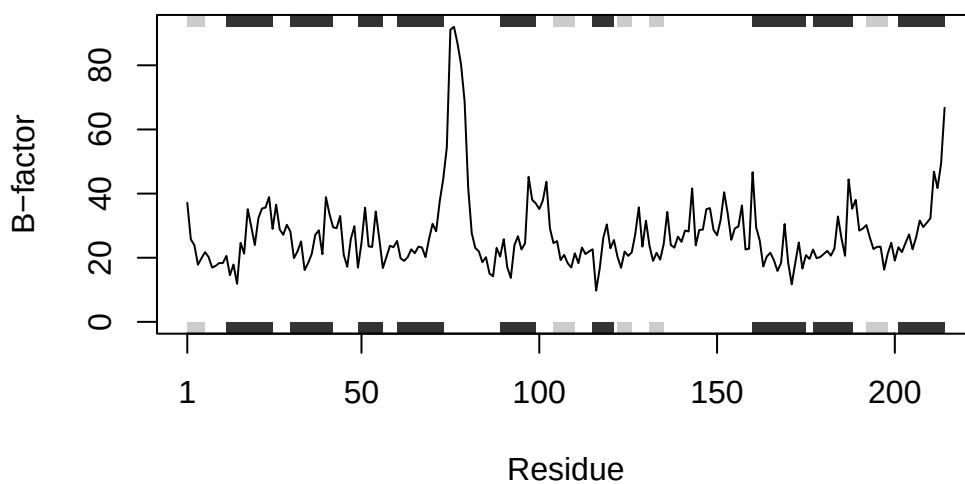
[1]	29.02	18.44	16.20	19.67	20.26	20.55	17.05	22.13	26.71	33.05
[11]	30.66	32.73	25.61	33.19	41.03	24.09	16.18	19.14	29.19	14.79
[21]	19.63	28.54	27.49	32.56	17.13	15.50	6.98	24.07	24.00	23.94
[31]	30.70	24.70	32.84	34.60	33.01	44.60	50.74	57.32	47.04	67.13
[41]	81.04	75.20	59.68	55.63	45.12	39.04	44.31	38.21	43.70	44.19
[51]	47.00	48.67	41.54	50.22	45.07	49.77	52.04	44.82	39.75	35.79
[61]	38.92	37.93	27.18	26.86	27.53	31.16	27.08	23.03	28.12	24.78
[71]	24.22	18.69	40.67	38.08	55.26	46.29	26.25	37.14	27.50	16.86
[81]	27.76	19.27	22.22	26.70	25.52	21.22	15.90	15.84	22.44	19.61
[91]	21.23	21.79	17.64	22.19	22.73	16.80	23.25	35.95	24.42	20.96
[101]	20.00	25.99	24.39	17.19	12.16	17.35	24.97	14.08	22.01	22.26
[111]	22.78	27.47	30.49	32.02	20.90	27.03	23.84	44.37	42.47	33.48
[121]	44.56	56.67	60.18	66.62	59.95	70.81	88.63	100.11	86.60	85.80
[131]	77.48	68.13	52.66	45.34	52.43	60.90	62.64	72.19	66.75	58.73
[141]	74.57	79.29	79.53	76.58	66.40	64.76	70.48	74.84	70.11	74.82
[151]	78.61	78.24	66.70	66.10	67.01	72.28	80.64	68.54	43.23	51.24
[161]	45.72	61.60	45.61	42.57	41.03	41.02	33.34	19.48	34.38	33.11
[171]	25.48	29.68	40.71	32.91	24.41	19.20	15.43	19.93	20.66	12.72
[181]	21.40	18.21	26.68	34.50	25.77	26.52	36.85	31.05	39.84	48.03
[191]	23.04	29.57	23.00	23.80	26.59	25.49	23.25	19.89	32.37	30.97
[201]	42.16	29.64	29.69	33.15	26.38	23.17	29.35	32.80	25.92	38.01
[211]	45.95	44.26	44.35	70.26						

```
analyze_protein("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

B-factor Plot for 1AKE

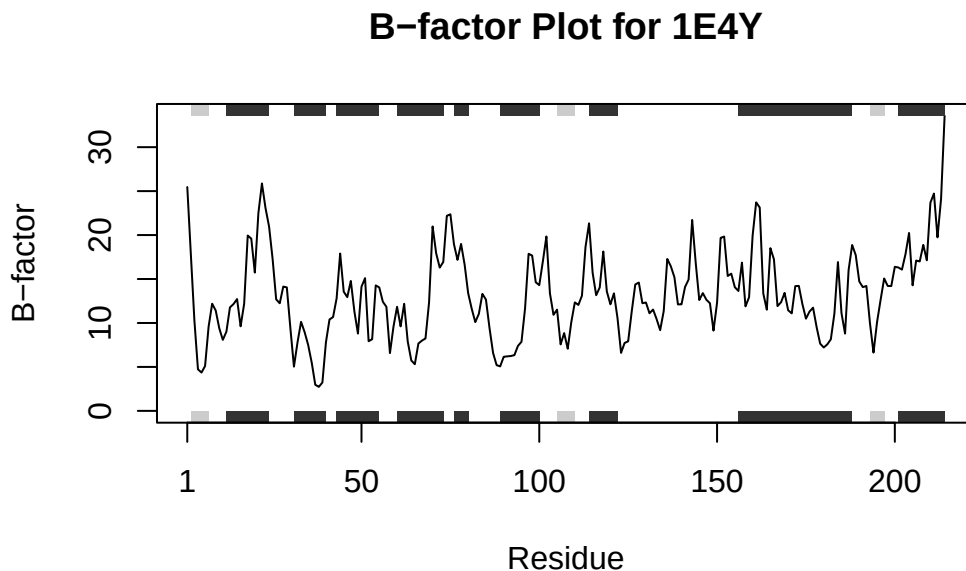


[1]	37.14	25.76	23.90	17.83	19.86	21.75	20.21	16.92	17.47	18.35	18.31	20.57
[13]	14.56	17.87	11.87	24.63	21.29	35.13	29.68	23.96	32.34	35.34	35.64	38.91
[25]	29.00	36.55	28.83	27.15	30.28	28.13	19.90	21.95	25.07	16.15	18.35	21.19
[37]	27.13	28.55	21.10	38.88	33.63	29.51	29.21	33.01	20.92	17.17	25.84	29.80
[49]	16.89	24.66	35.62	23.52	23.37	34.41	25.96	16.79	20.20	23.72	23.29	25.23
[61]	19.81	19.00	20.21	22.62	21.40	23.47	23.20	20.21	25.90	30.58	28.25	37.60
[73]	44.66	54.46	91.10	92.02	86.85	80.21	68.72	42.01	27.69	23.06	21.98	18.60
[85]	20.17	15.06	14.20	23.07	20.36	25.76	17.02	13.71	23.88	26.72	22.58	24.51
[97]	45.23	38.07	36.97	35.17	37.83	43.69	29.14	24.56	25.20	19.27	20.88	18.27
[109]	16.96	21.38	18.33	23.18	21.15	21.97	22.63	9.74	16.71	26.18	30.39	22.95
[121]	25.51	20.28	16.86	21.94	20.59	21.64	27.42	35.72	23.47	31.57	23.71	19.01
[133]	21.52	19.40	24.32	34.28	23.96	23.14	26.60	24.94	28.49	28.18	41.64	23.85
[145]	28.67	28.76	35.16	35.46	28.74	26.99	31.74	40.41	33.73	25.57	29.13	29.74
[157]	36.32	22.58	22.82	46.67	29.44	25.40	17.27	20.38	21.55	19.19	15.89	18.37
[169]	30.51	18.47	11.70	18.45	24.75	16.63	20.80	19.62	22.56	19.87	20.22	21.16
[181]	22.13	20.66	22.82	32.86	26.04	20.60	44.44	35.28	38.03	28.46	29.10	30.19

[193] 26.17 22.71 23.39 23.44 16.27 21.26 24.67 19.12 23.26 21.75 24.59 27.26
 [205] 22.63 26.40 31.60 29.57 30.90 32.29 46.86 41.73 49.31 66.76

```
analyze_protein("1E4Y")
```

Note: Accessing on-line PDB file



[1]	25.46	17.86	10.28	4.73	4.36	5.10	9.59	12.19	11.41	9.39	8.08	9.01
[13]	11.77	12.15	12.72	9.62	12.18	19.95	19.59	15.73	22.51	25.87	23.08	20.97
[25]	17.28	12.69	12.24	14.14	14.05	9.38	5.03	7.78	10.13	8.96	7.50	5.48
[37]	2.97	2.73	3.23	7.81	10.40	10.67	12.79	17.90	13.56	12.94	14.78	11.31
[49]	8.79	14.13	15.10	7.92	8.15	14.28	14.04	12.42	11.84	6.57	9.59	11.84
[61]	9.61	12.18	7.89	5.74	5.31	7.67	7.99	8.24	12.34	20.98	17.93	16.30
[73]	16.94	22.19	22.36	18.96	17.18	18.99	16.65	13.39	11.61	10.10	11.03	13.31
[85]	12.66	9.44	6.60	5.20	5.06	6.16	6.20	6.24	6.34	7.39	7.86	11.66
[97]	17.87	17.67	14.63	14.30	16.98	19.84	13.36	10.93	11.52	7.56	8.85	7.07
[109]	10.08	12.34	12.05	13.10	18.63	21.34	15.73	13.16	14.04	18.13	13.59	12.12
[121]	13.37	10.57	6.60	7.73	7.91	11.31	14.38	14.60	12.25	12.33	11.10	11.53
[133]	10.44	9.18	11.36	17.28	16.45	15.21	12.11	12.12	14.10	14.94	21.72	16.82
[145]	12.61	13.40	12.64	12.24	9.13	12.31	19.68	19.83	15.34	15.61	14.07	13.64
[157]	16.87	11.89	12.92	19.93	23.72	23.13	13.35	11.51	18.51	17.24	11.92	12.36

[169]	13.42	11.45	11.09	14.19	14.22	12.15	10.49	11.29	11.74	9.53	7.65	7.21
[181]	7.56	8.14	11.07	16.93	11.12	8.79	16.03	18.87	17.72	14.72	14.08	14.21
[193]	9.99	6.63	10.11	12.64	15.06	14.21	14.20	16.39	16.31	16.07	17.83	20.24
[205]	14.28	17.10	17.00	18.88	17.13	23.68	24.72	19.74	24.12	33.57		

The function takes in a PDB ID and an optional chain ID as inputs. It loads the protein structure, keeps only the alpha carbon atoms for the chosen chain, pulls out the B-factor values, and then makes a plot of them. It returns both the plot and the B-factor values as a numeric vector. The code is efficient since the same steps are not repeated for each protein, and everything is handled in one function. It is also easy to follow because the variable names make sense and the steps are in a clear order. When tested with valid PDB IDs, the function runs without issues and gives the expected results.